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Unit 1

1 **JavaScript Basics & Control Structures**
JavaScript basics include variables, conditions, and loops used to control the flow of a program.

2 **Functions, Arrays, and Events in JavaScript**
Functions organize code, arrays store multiple values, and events handle user interactions on web pages.

3 **ES6 Features: Classes, Arrow Functions, and Promises**
ES6 introduces modern features like classes, arrow functions, and promises for cleaner and asynchronous JavaScript.

4 **Express.js Basics and Application Setup**
Express.js is a Node.js framework used to create and run server-side web applications easily.

5 **Routing, Templates, and Error Handling in Express.js**
Express.js routing manages URLs, templates display dynamic views, and error handling manages application errors.

Unit 2

1. Hello World Program in ReactJS

A Hello World program in ReactJS displays a simple message on the screen using a basic component.

2. Creating Components

Components can be created as functions or classes to return JSX code for rendering UI.

3 Creating the First React App

The first React app is created using npx create-react-app or Vite and runs on a local development server.

4 JSX Programs

JSX programs combine HTML structure with JavaScript logic to create dynamic user interfaces.

5. ReactJS Props and Destructuring Props and State

State is a built-in object used to store and manage dynamic data inside a component.
Destructuring is a method of extracting values from props and state objects for cleaner code.

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Unit 3

1 Form Handling in React
Form handling in React manages user input from form elements like text fields, checkboxes, and buttons using component state and event handlers.

2 Event Handling and Binding Event Handlers
Event handling in React is used to respond to user actions such as clicks or typing using events like `onClick` and `onChange`. Event binding ensures the correct context of `this` when handling events inside components.

3 Conditional Rendering and List Rendering
Conditional rendering allows components to display different content based on conditions. List rendering displays multiple elements by iterating through arrays using methods like `map()`.

4 List, Keys, and Index as Key Anti-pattern
Keys are unique identifiers used when rendering lists in React to help efficiently update the UI. Using the array index as a key is considered an anti-pattern because it can cause rendering issues when list items change order.

5 React Components Lifecycle Methods, Pure Components, and Fragments
Lifecycle methods allow developers to control component behavior during mounting and updating phases. Pure components improve performance by preventing unnecessary re-renders, and fragments allow grouping multiple elements without adding extra DOM nodes.

Unit 4

1 React Memo
React Memo is a performance optimization technique that prevents unnecessary re-rendering of functional components by memoizing the component output when the props have not changed.

2 Refs in React (Refs with Class Components)
Refs in React provide a way to directly access DOM elements or React components. In class components, refs are created using `React.createRef()` and are commonly used to focus input fields or interact with DOM elements.

3 Forwarding Refs and Portals
Forwarding refs allows a parent component to pass a ref to a child component so it can access a DOM element inside the child. Portals allow rendering components outside the main DOM hierarchy while maintaining the React component structure.

4 Higher Order Components (HOC)
A Higher Order Component is an advanced React pattern used to reuse component logic. It is a function that takes a component as an argument and returns a new enhanced component with additional functionality.

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5 Rendering Props and Context in React
Rendering props is a technique where a component shares code by passing a function as a prop to another component. Context is used to share data across multiple components without passing props manually at every level of the component tree.

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6 HTTP and React (GET and POST Requests)
HTTP methods such as GET and POST are used in React applications to communicate with servers. GET requests are used to retrieve data from an API, while POST requests are used to send data from the client to the server.

Unit 5

1 Introduction to React Hooks and useState Hook
React Hooks are functions that allow developers to use state and other React features in functional components. The `useState` hook is used to manage state in a component and allows updating values dynamically during user interaction.

2 useState with Previous State, Object, and Array
The `useState` hook can update state based on the previous state value and can also manage complex data structures like objects and arrays. It helps in handling dynamic data changes in React applications.

3 useEffect Hook and Side Effects
The `useEffect` hook is used to perform side effects in functional components such as fetching data, updating the DOM, or setting timers. It runs after rendering and can be controlled to run conditionally, once, or with cleanup functions.

4 Fetching Data with useEffect and useContext Hook
The `useEffect` hook is commonly used to fetch data from APIs after the component renders. The `useContext` hook allows components to access shared data from a context without passing props through multiple component levels.

5 useReducer Hook and State Management
The `useReducer` hook is used for managing complex state logic in React applications. It works with a reducer function and actions to update the state and is useful when multiple state transitions are involved. It can also be compared with `useState` for handling simple versus complex state management.